

Linux Driver for Temperature and Humidity Sensor with HDC3022

Christopher Amadra Titus

December 10, 2025

Contents

1 Project data	1
2 Project description	1
2.1 Design and implementation	2
3 Project outcomes	2
3.1 Concrete outcomes	2
3.2 Learning outcomes	2
3.3 Existing knowledge	2
3.4 Problems encountered	2
4 Honor Pledge	2

NOTE: IT IS MANDATORY TO FILL IN ALL THE SECTIONS AND SUBSECTIONS IN THIS TEMPLATE

1 Project data

- Vittorio Zaccaria
- Describe in this table the group that is delivering this project:

Last and first name	Person code	Email address
Titus Christopher Amadra	11170417	christopheramadra.titus@mail.polimi.it

- Describe here how development tasks have been subdivided among members of the group, e.g.:
 - Foo worked on the overall threading infrastructure
 - Bar worked on lockfree access etc..
- Links to the project source code; Put here, if available, links to public repos hosting your project

2 Project description

2 pages max please

- What is your project about?
- Why it is important for the AOS course?

For those who choose to work on an open source project, please put here any reference/copy of messages exchanged with project maintainers to **identify the subject of the pull request**.

2.1 Design and implementation

Describe here the structure of the solution you devised. Note, don't put major parts of the source code here; if you can, put hyperlinks to existing repos.

For those who choose to work on an open source project, please put here an history (mail messages/github issues etc..) of the interaction with the development team that helped you identify such design and the code reviews that helped you improve it.

Those that have chosen to present a paper can exclude this section.

3 Project outcomes

3.1 Concrete outcomes

Describe the artifacts you've produced, if possible by linking to repo commits. For those who choose to work on an open source project, please put here the **URL to your final pull request**.

Those that have chosen to present a paper can include a link to the slides.

3.2 Learning outcomes

What was the most important thing all the members have learned while developing this part of the project, what questions remained unanswered, how you will use what you've learned in your everyday life? Please also indicate which tools you learned to use.

Examples:

- Foo learned to write multithreaded applications, he's probably going to create his own startup with what she has learned. She also learned how to debug with gdb.
- Bar learned how to interact with the open source community, politely answering to code reviews and issuing pull requests through Git.

3.3 Existing knowledge

What courses you have followed (not only AOS) did help you in doing this project and why? Do you have any suggestions on improving the AOS course with topics that would have made it easier for you?

3.4 Problems encountered

What were the most important problems and issues you encountered? Did you ever encountered them before?

- Foo encountered a problem with some critical sections. He ended up rewriting existing lock implementation.

4 Honor Pledge

(This part cannot be modified and it is mandatory to sign it)

I/We pledge that this work was fully and wholly completed within the criteria established for academic integrity by Politecnico di Milano (Code of Ethics and Conduct) and represents my/our original production, unless otherwise cited.

I/We also understand that this project, if successfully graded, will fulfill part B requirement of the Advanced Operating System course and that it will be considered valid up until the AOS exam of Sept. 2022.

Group Students' signatures